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Research note

Palaeomagnetism of quartz-magnetite rocks, Karimnagar, India

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Summary. Palaeomagnetic studies are reported for Precambrian (2590 ± 40 Myr) quartz-magnetite rocks from three localities of the Karimnagar area, Andhra Pradesh, India. Two localities yielded remanence directions of $D = 225^\circ 28'$; $I = -44^\circ 26'$ and $D = 225^\circ 31'$; $I = -41^\circ 28'$. The Virtual Geomagnetic Pole positions corresponding to these directions coincide with poles from the charnockites (about 2600 Myr) of Visakhapatnam, Eastern Ghats.

Introduction

The results of palaeomagnetic work on Indian rocks, reviewed in some detail by Athavale *et al.* (1970), and Bhimasankaram & Pal (1970), are mostly confined to a rather small span of time from the Upper Cretaceous to Miocene. Since palaeomagnetic results may provide clues to the history of the geomagnetic field, the mechanism of the continental drift and the age of the rock formations under study, the importance of systematic palaeomagnetic investigations on Precambrian geological formations cannot be underestimated. The results of a palaeomagnetic study on quartz-magnetite rocks from three localities; Manjampalli (18° 22′N; 79° 12′E), Kistaraopalli (18° 29′N; 79° 04′E) and Makdampur (18° 29′N; 79° 12′E) in Karimnagar district, Andhra Pradesh in India are reported in this paper.

Geology

The quartz-magnetite rocks selected for the present investigation are located at three villages as mentioned above. These rocks are in the form of small conical hillocks in the granitic terrain, with which they are associated and show conformable relations. They are of limited dimensions and occur purely locally pinching out at depth. At Manjampalli they are 25 m wide and run for about 1 km exhibiting an arcuate shape and outcropping discontinuously. Near Makdampur these rocks form an asymmetrical syncline about 100 m wide and 500 m in length. The quartz-magnetites at Kistaraopalli appear as a conical hill.

The age of quartz-magnetites has not been directly determined, but the age of granite gneisses was estimated to be 2590 ± 40 Myr by the total-rock Rb–Sr method using the

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decay constant of $1.39 \times 10^{-11} yr^{-1}$ for ^{87}Rb (Crawford 1969). Since the quartz-magnetite rocks are closely associated with granite gneisses and suffered the same deformation as in the case of granites, the quartz-magnetite rocks must also belong to the same age (Subba Rao 1974).

Three axial plane traces were observed in the folds of these rocks and suggest three phases of deformation and metamorphic activity in them. Petrographic studies indicated the presence of orthopyroxene as a primary metamorphic mineral, and secondary development of garnet and amphibole around ferrohypersthene margins and in the crushed zones (Subba Rao 1974). These lead to the interpretation that both the original ferruginous quartzites and granites were deposited together, and were involved in the initial stage of tectonism and metamorphism probably at 2590 Myr giving rise to quartz-magnetites and explaining the presence of orthopyroxene. The development of garnet and amphibole was attributed to the second phase of metamorphism and structural disturbance. It was observed that the slip planes in quartz-magnetites were occasionally filled up with pure and coarse magnetite. This suggests that the primary magnetite of the quartz-magnetites had been remobilized and recrystallized during the second phase, which must be very intensive. The third event was characterized by the absence of any visible alteration and this is positively due to its lesser intensity.

Sampling

The samples used in this present investigation are classified according to their location. The first letter corresponds to the locality, the following number to the sample and the number within brackets refers to the number of the specimen cut from the same sample. Mj thus stands for Manjampalli, M for Makdampur and K for Kistaraopalli. At least two specimens were drilled from each sample and a total of 31 specimens was prepared and used in the present investigation.

NRM and demagnetization studies

The natural remanent magnetism of the above specimens was determined using an astatic magnetometer. Fig. 1 shows the directions of NRM of the specimens sitewise. All these specimens were also subjected to af demagnetization tests and the directions after cleaning were determined. The specimens were subjected to progressively increasing alternating fields from 25 to 800 Oe and the intensity of magnetization J and its direction were determined at each field. The procedure adopted is essentially the same followed by Creer (1959). The normalized intensities J/J_0 and the directions of magnetization at different af demagnetization fields are shown in Fig. 2 for a few representative specimens. It is observed that a peak field of 600 Oe is sufficient to eliminate the secondary components in the rocks of Kistaraopalli and Makdampur. The directions of remanent magnetism, when the specimens were cleaned in an alternating magnetic field of 600 Oe, are shown in Fig. 1. Fig. 2 further shows that the direction of magnetization in the specimens from these two areas undergoes considerable variation at lower fields, while the change at higher fields of 600–800 Oe is small. For specimens from Manjampalli, however, peak fields of even 800 Oe are insufficient to remove the secondary components.

Thermo-magnetic studies were conducted on a few specimens and representative thermo-magnetic curves are shown in Fig. 3. The Curie temperatures for these three samples (namely

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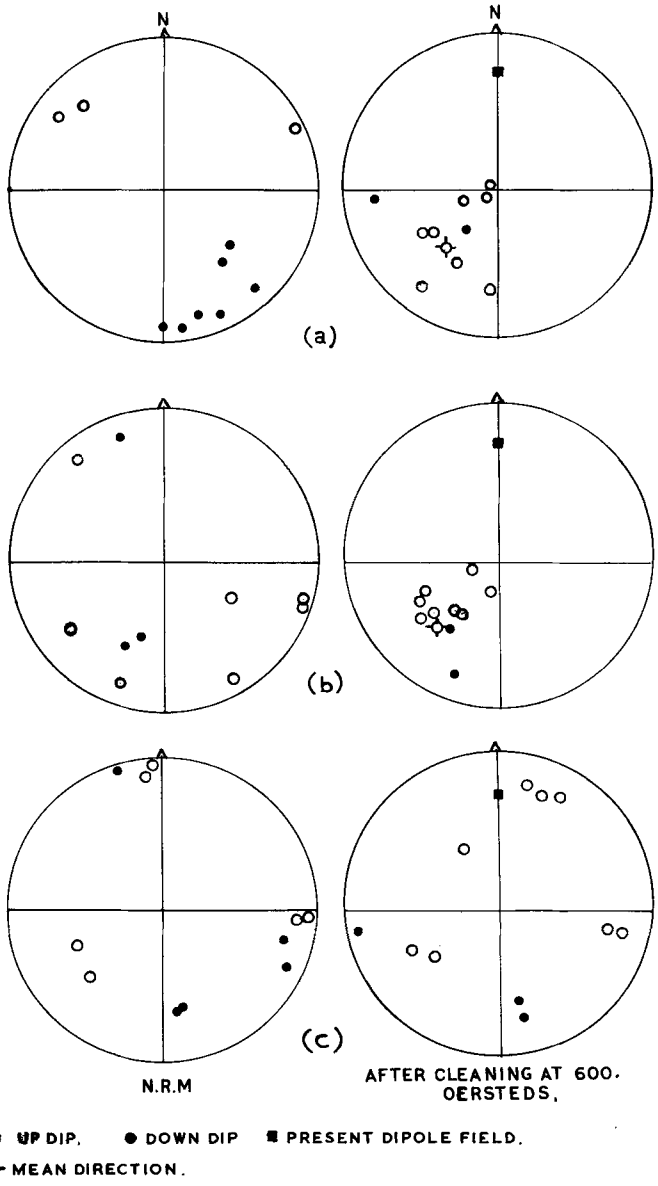


Figure 1. Equal area projection of directions of magnetization. (a) Kistaraopalli area, (b) Makdampur area, (c) Manjampalli area.

Mj2, K3, M3) are 604, 596 and 602°C respectively. Note also the reversible character of the saturation magnetization curves in Fig. 3. Irving (1963) notes that specimens with Curie temperatures less than 540°C tend to be less stable and concludes that rocks with stable magnetism show reversible character on cooling.

Results and discussion

The mean directions of magnetization of the quartz-magnetites for each site after af cleaning and the corresponding palaeomagnetic pole positions for each site, as obtained from Fisher's

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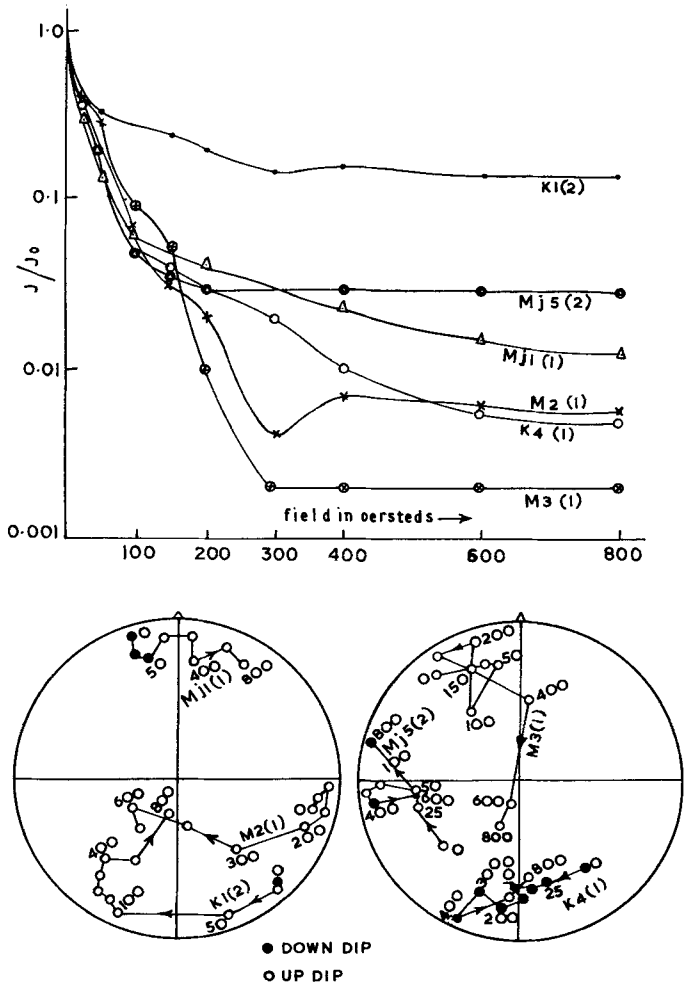


Figure 2. Changes of directions of magnetization in various peak field strengths with their corresponding normalized intensities of some representative specimens.

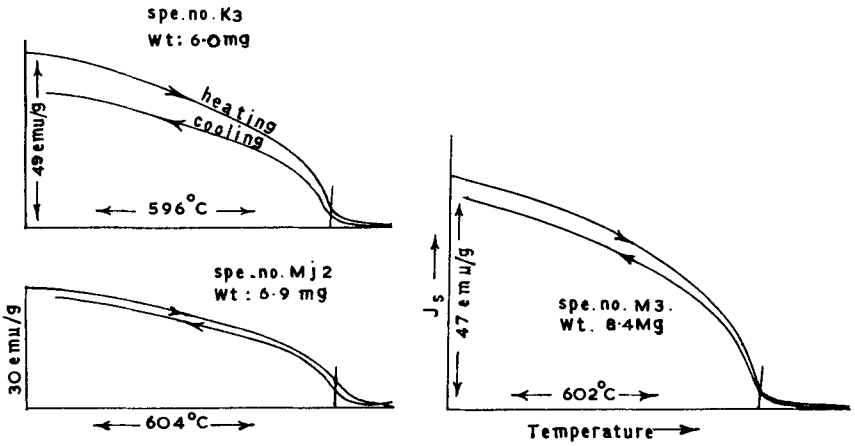


Figure 3. Thermo-magnetic curves of some representative specimens.

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Table 1. Mean directions and pole positions of quartz-magnetites.

Site	Location of site		N	Mean direction of the remanent magnetism after af cleaning		R	k	α_{95}	Palaeomagnetic pole	
	Latitude	Longitude		D	I				Latitude	Longitude
Kistaraopalli	18° 29′ N	79° 04′ E	10	225° 28′	−44° 26′	7.61	3.76	23.4°	47° 27′ S	29° 48′ W
Makdampur	18° 29′ N	79° 12′ E	10	225° 31′	−41° 28′	7.80	4.08	22.5°	47° 24′ S	26° 11′ W

N = Number of specimens. D = Declination of the remanent magnetic vector. I = Dip of the remanent magnetic vector. R = Length of the resultant magnetic vector. k = Precision parameter. α_{95} = Radius of 95 per cent confidence circle.

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(1953) analysis, are shown in Table 1. While calculating the pole positions and R, k and α_{95} values, all the data were made use of including the positive dips. This explains the relatively higher values of α_{95}, in spite of the satisfactory grouping in Fig. 1. The pole positions are in close agreement with the results of charnockites of Visakhapatnam (Bhimasankaram 1964) which are assigned an age of 2600 Myr from the age determinations by Vinogradov *et al.* (1964) using the lead-isochron method and Pb^{207}/Pb^{206} isotope ratio; these charnockites are part of the Eastern Ghats of India believed to have been emplaced at about 2600 Myr. They were later subjected to a strong metamorphic activity giving rise to palingenetic charnockites, pegmatites and mineralized zones (Rao & Babu 1978) at 2000 ± 100 Myr (Vinogradov *et al.* 1964). The palaeomagnetic age of the charnokites may therefore be assigned as 2000 Myr which is the age of second metamorphic activity in these rocks.

The Eastern Ghats of India are believed to have been subjected to three stages of metamorphic activity at ages of 2600, 2000 and 500 Myr respectively (Sarkar 1968; Crawford 1974), the second phase being the strongest. The metamorphic activity at 500 Myr is believed to be of minor intensity as it was characterized with the uplift of the Eastern Ghats only and not with the alteration to or formation of new minerals; we believe that it is unlikely to have achieved significant remagnetization.

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References

Athavale, R. N., Verma, R. K., Bhalla, M. S. & Pullaiah, G., 1970. Drift of the Indian sub-continent since Pre-cambrian times, *Palaeogeophysics*, p. 291, Proceedings of NATO, Advanced Study Institute, Academic Press.

Bhimasankaram, V. L. S., 1964. A preliminary investigation on the palaeomagnetic directions of the charnockites of Andhra, *Pradesh. Curr. Sci.*, **33**, 465.

Bhimasankaram, V. L. S. & Pal, P. C., 1970. Palaeomagnetic studies in India – a review, *Proc. Second Symp. on Upper Mantle Project*, p. 223, 1970 December, Hyderabad.

Crawford, A. R., 1969. India, Ceylon and Pakistan, new age data and comparisons with India, *Nature*, **223**, 380.

Crawford, A. R., 1974. Indo-Antarctica, Gondwanaland, and the distortion of a Granulite belt, *Tectonophys.*, **22**, 141.

Creer, K. M., 1959. A. C. demagnetisation of unstable Triassic Keuper Marls from S.W. England, *Geophys. J.*, **2**, 261.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. London A*, **217**, 295.

Irving, E., 1963. The magnetism of some Permian rocks from New South Wales, *Geophys. J.*, **7**, 395.

Rao, A. T. & Babu, V. R. R. M., 1978. Allanite in Charnockites from Air Port Hill, Visakhapatnam, Andhra Pradesh, India, *Am. Mineral.*, **63**, 330.

Sarkar, S. N., 1968. *Precambrian Stratigraphy and Geochronology of Peninsular India*, Dhanbad Publishers, India.

Subba Rao, Y. V., 1974. Some geological and geophysical investigations around Karimnagar Town, Andhra Pradesh, *PhD thesis* (unpublished), Andhra University, Waltair.

Vinogradov, A., Tugarinov, A., Zhykov, C., Stapnikova, N., Bibikova, E. & Khorre, K., 1964. Geochronology of Indian Precambrian, *Proc. 22nd Int. Geol. Congr.*, **10**, 553.

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